

Elementary Paper Animatronics

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Figure 1: A sample paper animatronics project. Elle the Elk explains the different winter holiday celebrations in Toronto.

ABSTRACT

Paper Animatronics is a new way for elementary school kids to engage with subject matter through project-based learning. We're upgrading the classic shoe box diorama, and empowering kids to bring it to life by adding servo motors, sound and lights to create compelling characters and shows. In this workshop designed for teachers, parents and other advocates for creative education, you will get to work hands-on with our new paper animatronics kits.

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These make it easy to create talking characters that you can voice in real-time or use in more complex, scripted shows where things move and light up on cue using a synchronized Arduino program. Through these activities, you will see how kids learn to be creative across both technical and artistic disciplines as they explore class subject matter.

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1 PAPER ANIMATRONICS

Paper Animatronics is a project-based learning activity where elementary school create characters and stories and bring them to life through papercraft with sound and motion! Like making posters or dioramas, paper animatronics can be used to reinforce learning in almost any subject.

To ease construction, we have developed unique mechanical components such as our *Zip Servos* (Figure 2). These can be taped to a piece of cardboard and provide simple linear motion. Figure 3 shows a simple example linear mouth movement.

Our programs for elementary schools have three levels of activities targeted for early, middle and late elementary school students, allowing them to build their skills over time.

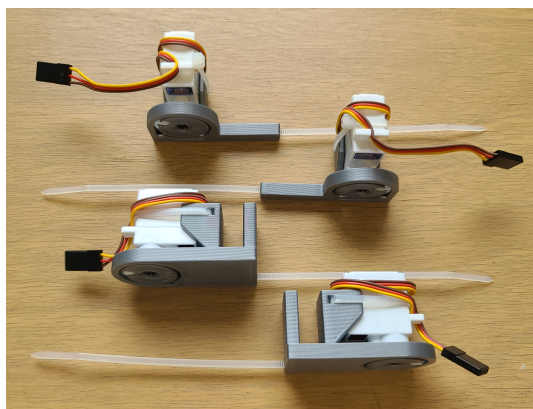


Figure 2: The four varieties of Zip Servos that provide simple linear motion.



Figure 3: This show uses Zip Servos to give an impressive range of motion.

1.1 Voice Puppetry (Early Elementary)

In this activity, the child creates a simple paper character with a moving mouth, and attaches a motor. The motor is then connected to our Mic2Servo board (Figure 4) that moves the mouth up and down in real-time as the child speaks.

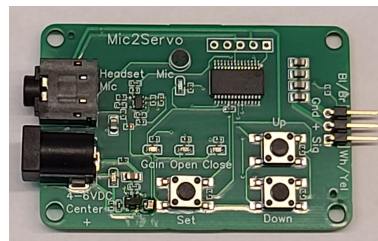


Figure 4: The Mic2Servo board has a microphone and drives a servo to create appropriate mouth motions.

1.2 Two-Character, Scripted Shows (Middle Elementary)

In this second activity, story and script take center stage as kids create a show with two characters talking to each other. Similar to the first activity, kids create two paper characters with moveable mouths connected to motors. They write a script for the two characters, and then use audio editing tools like Audacity to record and process (pitch shifting and other effects) their own voices. The resulting stereo audio file is then used to drive the two motors via our Audio2Servo board (Figure 5).

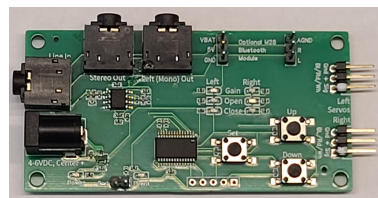


Figure 5: The Audio2Servo board takes a stereo audio signal, and drives two mouth servos.

1.3 Synchronized Shows (Late Elementary)

In the third activity, we augment our paper shows using a microcontroller (e.g. Arduino), allowing a wide range of synchronized motions, lights and outputs. This provides a purposeful, gentle introduction to programming using a simple state machine. The Audio2Servo board includes a logic output which provides synchronization signals triggered by audio on the right channel.

2 WORKSHOP

In our SIGGRAPH hands-on workshop, participants will create a simple paper character and make its mouth move with a servo. This will first be tested using voice puppetry via our Mic2Servo board.

The second activity in the workshop will use our Audio2Servo board to create a synchronized show featuring the previously crafted character. Participants will use Audacity on a PC to create a stereo audio file, with the left channel holding the character speech, and the right channel holding synchronization sounds. These sounds will be used to trigger an Arduino to provide additional synchronized lights and motion.